1. SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

1. **REQUIREMENTS ANALYSIS**

- **Goal:** Understand what the community needs.
- **Activities:**
    - Gather input from donors, beneficiaries, and pantry managers.
    - Define system requirements — e.g., donation tracking, inventory management, beneficiary registration.
    - Identify constraints like internet access or data privacy.
- **Output:** Requirement Specification Document.

2. **SYSTEM DESIGN**

- **Goal:** Plan how the system will work.
- **Activities:**
    - Design the architecture (frontend, backend, database).
    - Create data flow diagrams showing how donations move through the system.
    - Define user interfaces for donors, managers, and beneficiaries.
- **Output:** System Architecture and Design Document.

3. **IMPLEMENTATION (CODING)**

- **Goal:** Build the actual software.
- **Activities:**
    - Develop modules: Donor Interface, Inventory Database, Beneficiary Portal, Notification System.
    - Use structured programming principles for clarity and maintainability.
    - Integrate backend logic with the database.
- **Output:** Working prototype or application.

4. **TESTING**

- **Goal:** Ensure the system works correctly.
- **Activities:**
    - Unit testing for each module.
    - Integration testing to verify data flow between components.
    - User acceptance testing with real community members.
- **Output:** Tested and validated system ready for deployment.

5. **DEPLOYMENT**

- **Goal:** Make the system available for use.

- **Activities:**
  - Host the system online or on local servers.
  - Train pantry managers and volunteers.
  - Launch pilot testing in selected communities.
- **Output:** Operational Digital Community Pantry Connector.

6. **MAINTENANCE**

- **Goal:** Keep the system running smoothly.
- **Activities:**
  - Fix bugs and update features.
  - Monitor performance and user feedback.
  - Add new functionalities (e.g., analytics dashboard or mobile app).
- **Output:** Updated and optimized system.

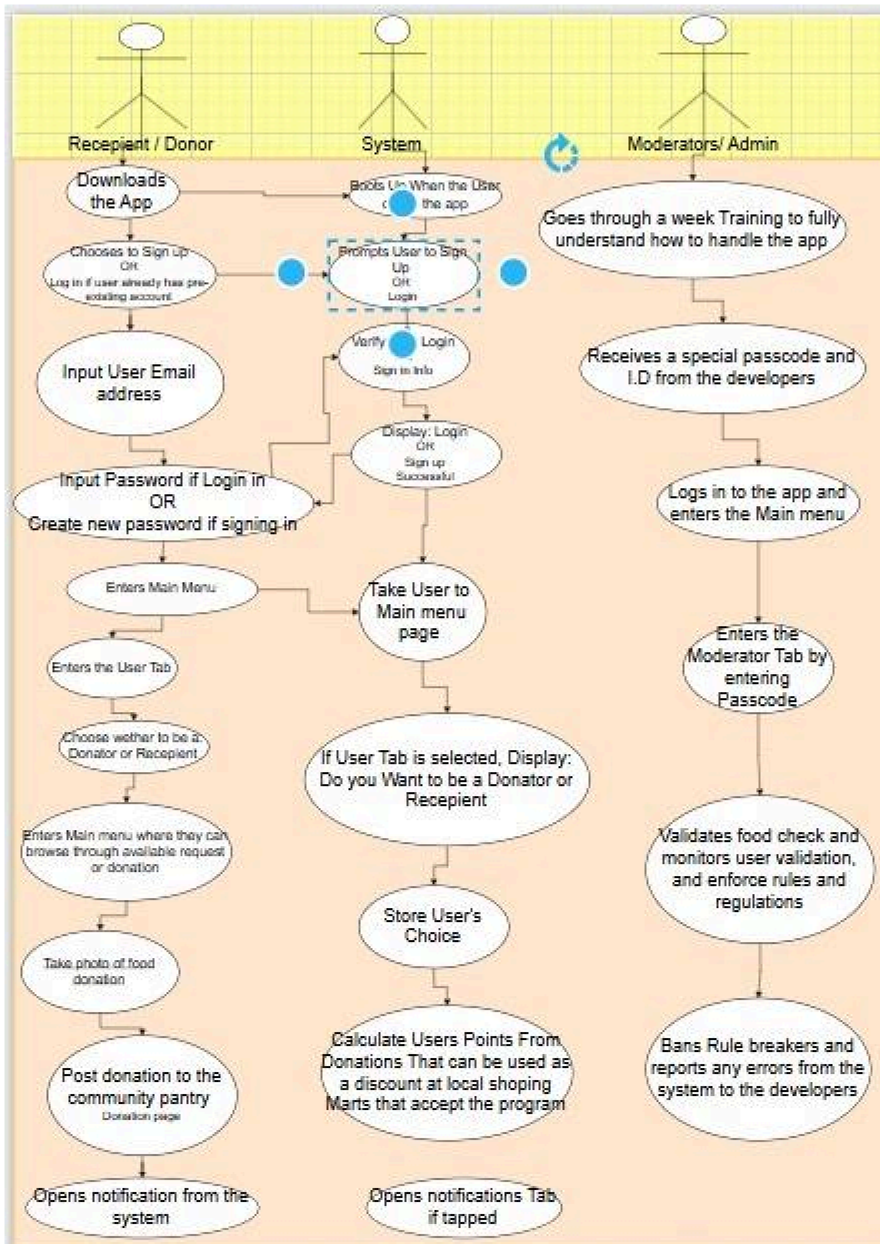
Interaction Across Phases

Phase	**Key Component**	**Interaction**
Requirement	Donor & Manager	Define donation categories
Design	Database & UI	Map item flow from donor to beneficiary
Implementation	Backend & Notification	Trigger alerts when stock is low
Testing	All modules	Verify donation approval process

Deployment	Logistics	Enable pickup scheduling
Maintenance	Admin	Update expired item alerts

2. SYSTEM ARCHITECTURE DESIGN

Use case Diagram



1. How the components interact

In a digital community pantry connector, the components typically interact in a structured flow that ensures donations, inventory, and beneficiaries are coordinated efficiently.

Core Components and Interactions

1. **Donor Interface**
    - Donors register or log in.
    - They input available items (e.g., rice, canned goods).
    - Data flows into the **Donation Management System**.
2. **Donation Management System**
    - Validates donor entries (quantity, expiry dates).
    - Updates the **Inventory Database** with new stock.
    - Sends confirmation back to donors.
3. **Inventory Database**
    - Central repository of all pantry items.
    - Tracks stock levels, categories, and expiry dates.
    - Provides real-time data to both **Pantry Managers** and **Beneficiary Interface**.
4. **Pantry Manager Dashboard**
    - Monitors inventory levels.
    - Approves or rejects donations.
    - Allocates items to beneficiaries based on need.
    - Communicates with **Logistics/Distribution System**.
5. **Beneficiary Interface**
    - Beneficiaries register and request items.
    - Requests are matched against available inventory.
    - Approved requests trigger notifications.
6. **Logistics/Distribution System**
    - Coordinates pickup or delivery.
    - Updates status (e.g., “Ready for pickup”).
    - Sends confirmation to both **Pantry Managers** and **Beneficiaries**.
7. **Notification System**
    - Sends SMS, email, or app alerts to donors, managers, and beneficiaries.
    - Keeps all parties updated on donation status, approvals, and deliveries.

Interaction Flow Example

- **Donor → Donation Management → Inventory → Pantry Manager → Beneficiary → Logistics → Notification**
- Each step feeds into the next, ensuring transparency and accountability.

Project planner

ACTIVITY	PLAN START	PLAN DURATION IN MONTHS	ACTUAL START MONTH	ACTUAL DURATION	PERCENT COMPLETE
Gather input System	FEBRUARY	6	JUNE	4	30%
Requirements Identify	AUGUST	9	OCTOBER	10	75%
Constraints Front and	JULY	4	JULY	5	35%
backend	AUGUST	8	SEPTEMBER	6	10%
Coding	JUNE	12	JUNE	8	85%
Testing	JULY	7	JANUARY	6	85%
Deployment	FEBRUARY	4	MARCH	3	50%

SDG Link

The community Pantry Connector system primarily supports **SDG 2: Zero Hunger** and **SDG 12: Responsible Consumption and Production**. By acting as a digital bridge between surplus and need, it transforms potential waste into a community resource.

SDG 2: Zero Hunger

This goal aims to end hunger, achieve food security, and improve nutrition.

- **Target 2.1 (Universal Access):** The system provides a low-barrier way for vulnerable populations to access safe, nutritious food that might otherwise be unaffordable.
- **Direct Contribution:** By digitizing the "invisible" surplus in a neighborhood, the system ensures that edible food reaches people's plates instead of landfills, directly addressing local food insecurity.

SDG 12: Responsible Consumption and Production

This goal focuses on sustainable patterns of using resources.

- **Target 12.3 (Halve Food Waste):** The goal specifically calls for halving per capita global food waste at the retail and consumer levels by 2030.
- **Direct Contribution:** The app facilitates the **circular economy** by enabling the redistribution of unsold or extra items from households, bakeries, and cafes. This reduces the total volume of food discarded, helping hit the 50% reduction target.